



Time	14 Oct Wednesday - Neural Basis	15 Oct Thursday - AI and Cog.Sci.	16 Oct Friday - Robotics and Applications
8'30-8'55 8'55-9'05	Registration Auditorium CSIC <u>Opening</u> José María Martell Vice-president for Scientific and Technical Research (CSIC) <u>Opening</u> Pablo Lanillos and IWAI organizers	Doors open	Doors open
9'05-10'00	Keynote: Neural basis (45'+10') Prof. Rajesh Rao	Keynote: Artificial Intelligence (45'+10') Prof. Tadahiro Taniguchi. "Collective Predictive Coding Hypothesis: Emergence of Language and Cooperative Behaviors"	Keynote: Robotics (45'+10') Prof. Carme Torras "Healthcare robotics: research challenges, co-design, and technoethics education"
10'00-10:30	Coffee	Coffee	Coffee
10'30-11:30	Session 1 Predictive Coding and Representations Chair: Pablo Lanillos	Session 3 Mathematical Foundations of Active Inference Chair: Martin Wisse	Session 5 Multi-Agent, Collective Active Inference, and Robotics Chair: Hideaki Shimazaki
11'30-12:00	Spotlight Session 1	Spotlight Session 3	Spotlight Session 5
12'00-13'00	Fireside debate: Action! The brain representation of behavior Rajesh Rao (Washington), Liset de la Prida (CSIC), Ruben Moreno Bote (Pompeu Fabra), Karl Friston (UCL) <u>Moderator:</u> Ivilin Stoianov	Fireside debate: Ghost in the machine: AI-extended intelligence Roderick Murray-Smith (Glasgow), Sara Degli Esposti (CSIC), Karl Friston (UCL) <u>Moderator:</u> Pablo Lanillos	Fireside debate: Are robotaxis ready? Jorge Villagrà (CSIC), Adam Bahke (Motor.ai), Johan Engström (Waymo), Karl Friston (UCL) <u>Moderator:</u> Martijn Wisse.
13'00-14'00	Lunch	Lunch	Lunch
14'00-14'10	Sponsors presentations	Sponsors presentations	Sponsors presentations
14'10-15'10	Session 2 Motivation, Interoception, and Affect in Active Inference Chair: Mahault Albarracín	Keynote: Cog. Models (45'+10') Prof. Giovanni Pezzulo. "From Cognitive Maps to Schemas: Active Inference for Flexible Navigation"	Session 6 Language, Clinical Applications, and Perspectives Chair: Daria de Tingly
15'10-15'40	Spotlight Session 2	Spotlight Session 4	Spotlight Session 6
15'40-16:00	Coffee	Coffee	Coffee
16:00-17:00	Poster session 1	Session 4 Planning, Decision, and Control Chair: Ivilin Peev Stoianov	Poster session 3
17:00-18:00	Welcome	Poster session 2	Closing remarks
18:00-19:30	Welcome		
19:30-23:00		Networking dinner	



Contributed Talks

Session	ID	Title	Authors
1. Predictive Coding and Representations	59	Error Highways: Scaling Predictive Coding to Very Deep Networks	Amirhossein Mohammadi, Alex Ororbia
	60	Zero-Shot Representation Learning of Visual Sensory Stimuli through Laterally-Competitive Predictive Coding	Faezeh Habibi, Amirhossein Mohammadi, Alex Ororbia
	48	Sensory Precision Inference for Multimodal Arbitration under Uncertainty	Tin Mišić, Takato Horii
	73	Vision-Language Models as Gaussian Prior Preferences for Active Inference	Sensho Nobe
2. Motivation, Interoception, and Affect in Active Inference	36	Emotions as Deep Neuromorphic Priors in Self-Motivated Active Inference Agents	Elliot A. Giles, Paolo Pagliuca
	93	Two routes to interoceptive prioritisation: dissociating likelihood precision and preference dynamics in active inference	St John Grimbly
	97	Can Network Models of Science Do Without Pluralist Agents?	Jonas Hallgren, David Hyland, Mahault Albarracin
	23	Belief-Based Maximum Occupancy Principle and Active Inference	Manolis Mylonas, Rubén Moreno-Bote
3. Mathematical Foundations of Active Inference	84	Parameter Learning in Active Inference via Variational Message Passing	Benedikt Waldeck
	14	Expected free energy as an information constraint on the Bethe Lagrangian	Wouter M. Kouw
	21	Variational Laplace at Two Loops: Feynman Diagrams for the Variational Free Energy.	Stefano Goria
	46	Renormalising Generative Models for Active Inference: Foundations, Derivations, and Verification	Karim Zaghw, Andrew Pashea, Marc Pritsch, Wouter W. L. Nuijten, Karl Friston, Lancelot Da Costa
4. Planning, Decision, and Control	94	Active Inference as a Convex Markov Decision Process	Nikola Milosevic, Nicolás Hinrichs, Nico Scherf
	80	BeFree-Gate: Policy Composition under Partial Observability	Francesca Rossi, Giovanni Russo
	13	Sophisticated Policies from Epistemic Priors	Wouter W. L. Nuijten, Bert de Vries
	72	Active Inference with Quantum-like Cognitive Control	Catherine F. Higham
5. Multi-Agent, Collective Active Inference, and Robotics	6	Collective sensing as emergent bayesian inference	Conor Heins
	101	Expected Free Energy-based Informative Path Planning for Robotic Mars Exploration	Ajith Anil Meera, Pablo Lanillos, Wouter M. Kouw
	102	From Environmental Scaffolding to Internalized Beliefs: Resilient Leader–Follower Configurations in Active-Inference Collectives	Alejandro Jimenez Rodriguez, Pranay Vaddepalli, Mahault Albarracin
	35	Active Inference for Decentralised Traffic Coordination: A Multi-Agent Route-Choice Demonstration	Xue Yao, Dmitry Bagaev, Yanan Xin, Serge Hoogendoorn, Hans van Lint, Bert de Vries
6. Language, Applications, and Perspectives	113	One complex system organizing another: language signal structures hierarchical generative models across the lifespan	Evguenia Malaia
	18	Balancing time and precision with active sensor selection and positioning	Esther G. van Pelt, Alexandru Onose, Ruud van Sloun, Harm Belt, Wouter M. Kouw
	19	The clinician as affordance-architect: restoring breadth and depth of planning in persistent dizziness.	Marc A Broberg, Mick Thacker, Karl Friston, Jorge E Esteves
	67	Transfinite Anatomy Theory: Selective Hierarchical Boundaries for Disease Modeling	Leon Moriguchi



Spotlight Session	ID	Title	Authors
1. Predictive Coding and Reoresentations 14 Oct 11'30 - 12'00 Chair: Pablo Lanillos → Poster Session 1	5	Primitive Representation by a Low-Dimensional Deterministic Bias on Predictive-Coding-Inspired Variational RNN	Takahito Fujimori, Jun Tani
	50	Reverse engineering variational invariant structures in rodent brains	Takuya Isomura
	52	Single State Update Predictive Coding training for Time Series Forecasting and Anomaly Detection	Matteo Cardoni, Sam Leroux
	58	Exploring napping paradigm for Recurrent Spiking Neural Networks	Andreas Lie Massey Stefano Nichele, Aliaksandr Hubin
	75	A scale-invariant link between neural dynamics and active inference	Ukyo T. Tazawa, Ken Nakae, Teppei Ebina, Masanori Matsuzaki, Kenji Doya, Takuya Isomura
	79	Precision-Adaptive Spiking Free Energy Networks for Control	Javier Zaragozaano, André Rodrigues Urbano Sander Wessel Keemink, Pablo Lanillos
	122	Active, Actionable Scene Graphs	Pablo Bustos, Noé José Zapata Cornejo, Pedro Núñez
	74	When Do Scene Beliefs Help Hard Attention? A Two-Stage Scene-Map Intervention Study	Pengcheng Pan, Yukie Nagai
2. Motivation, Interoception, and Affect in Active Inference 14 Oct 15'10 - 15'40 Chair: Mahault Albarracin → Poster Session 1	11	Neural Correlates of Mutual Information Beyond Uncertainty in Curiosity-Driven Exploration	Takuma Miyaguchi, Shin Nishimura Kazutaka Ueda, Hideyoshi Yanagisawa
	38	Arbitration Between Competing Needs and the Emergence of Goal Commitment in Active Inference	Matteo Priorelli, Giovanni Pezzulo
	112	Partner-Specific Affective Precision in Social Active Inference	Harshil Shah, Andrew Pashea
	55	Emotion in an active inference model of human driving	Julian F. Schumann, Johan Engstrom, Ran Wei, Jens Kober, Martijn Wisse, Arkady Zgonnikov
	65	Conflict Resolution and Failure in Hierarchical Active Inference	Jaeseo Han
	29	The Searchlight Test Revisited: An Active Inference Model Reproduces Disorganized Visual Exploration in Schizophrenia	Kazuki Yasuda, Hayato Idei, Hiroki Kojima, Ryosuke Izawa, Masashi Ikeda, Yuichi Yamashita
	81	Cardiac Signals as Precision-Weighted Priors: An Active Inference Account of Heartbeat-Modulated Emotional Perception	Teresa Garzón Molina
	117	Principled Direction-Free Intrinsic Motivation through Model-Free Epistemic Free-Energy Estimators	Alireza Furutanpey, Schahram Dustdar
3. Mathematical Foundations of Active Inference 15 Oct 11'30 - 12'00 Chair: Martin Wisse → Poster Session 2	32	Dröste Hierarchies in Predictive Processing: A Scale-Invariant, Non-Well-Founded Model of Recursive Selfhood in the Bayesian Brain	Burak Erdeniz, Aybars Yıldırım VID
	39	Constructing Markov Blankets from Actions.	Joséphine Pazem; Hans J Briegel, Lukas J. Fiderer
	43	Physics of Information Geometry: Principle of Least Action on the Probability Simplex	Can Emre Koksal, Deniz Sargun
	4	Expected Free Energy as Belief-Dependent Utility for p-POMDPs	Patrick Cooper
	89	Distributional Active Inference	Abdullah Akgül, Gulcin Baykal, Manuel Haussmann Mustafa Mert Çelikok, Melih Kandemir
	49	Uniqueness Maximization Active Inference for Encoder-Only Robotic Perception and Behavior Learning	Viet Dung Nguyen; Alex Ororbia
	62	An On–Manifold Cost Repaid as Off–Manifold Robustness: Dissecting Information–Gain in Continuous–Control Active Inference	Stefano Gorla
	111	Where Value Lives: A Comparative Survey of Reinforcement Learning, Control as Inference, and Active Inference	Emma Graham, Chelsea Joe, Daniel Ari Friedman
4. Planning, Decision, and Control 15 Oct 15'10 - 15'40 Chair: Ivilin Stoianov → Poster Session 2	24	Future shapes present: autonomous goal-directed and sensory-focused mode switching in a Bayesian allostatic network model	Hayato Idei, Jun Tani Tetsuya Ogata, Yuichi Yamashita
	40	Active Inference for Interaction-Mediated Control of a High-dimensional Robotic Arm	Fraser Paterson, Sebastian Stein, Markus Klar John H. Williamson, Roderick Murray-Smith
	47	Implicit Leadership Transitions in Shared Teleoperation via Free-Energy-Based Meta-Prior Adaptation	Hiroki Sawada, Shunichi Kasahara
	66	Shared Goal Understanding in Physical Human-Robot Interaction through Deep Active Inference	Gabriel W. Haddon-Hill Ferdinand Hartmann, Shinao Murata Nika Hosseini
	95	Active Inference for Communication-Aware Update Selection in Interactive Holographic-Type Communication	Martin Maier, Zakaria Abou el Houda
	96	Benchmarking Generative Policy Proposals for Scalable Active-Inference Control	Yulin Li, Mohsen a jafari, Andrea Matta
	3	JAXbounce: A Continuous-Control Pong-Like Benchmark for Active Inference Beyond Low Dimensional Environments	Chaitanya Kaul, Markus Klar, Sebastian Stein, Fraser Paterson, John H. Williamson, Roderick Murray-Smith
	78	Precision-Weighted Attentional Orienting: A POMDP Model of the Posner Cueing Task with Eye-Tracking Data	Luna Macías Salgado
5. Multi-Agent, Collective Active Inference, and Robotics 16 Oct 11'30 - 12'00 Chair: Hideaki Shimazaki → Poster Session 3	10	VAMAI: Variational Affordance Mapping and Active Inference for Training-free Autonomous Drone Navigation	Satyaki Maitra, Harshil Shah
	25	Robotic Soccer Self-Localisation with active inference	Charel Van Hoof, Martijn Wisse, Noah van der Meer
	30	Expected Free Energy Planning with Learned External-Camera Measurement Reliability	Joost Leliveld, Omur Arslan, Wouter M. Kouw
	45	Deep Active Inference for Real-World Robotic Object Manipulation with Semantic Image Embeddings	Ko Igari, Kentaro Fujii, Shingo Murata
	53	More Than Reward: Uncertainty Aware Ranked Sampling in Affordance-Conditioned Autonomous Tool Design	Abián Torres, Pablo Lanillos, Ricardo Sanz
	71	Distributed Mine Search with Active Inference over Layout Hypotheses	Alexandre La Grappe, Daria de Tinguy, Emile Le Flecher, Joao Borges de Sousa, Yoshiyuki Nishio
	120	A Stochastic-Thermodynamic Active Inference Framework for Decentralized Swarm Control	Guido Quirijn Insinger
	99	Hierarchical Active Inference and Meta-Learned World Models for Cooperative Multi-Agent Games	Alejandro Jimenez Rodriguez Mahault Albarracin, Luca Manneschi
6. Language, Applications, and Perspectives 16 Oct 15'10 - 15'40 Chair: Daria de Tinguy → Poster Session 3	64	Asymmetric collaborative pricing via Active Inference and Mutual Modelling	Parth Patel, Kalliopi E Giannakopoulou, Ioanna Patsaroucha, Roderick Murray-Smith, Sebastian Stein
	8	Speech that Protects a Model: A Type-Safe Active-Inference Engine for Compensatory-Defensive Discourse	Maria Svet
	26	Hardware-Aware and High-Performance Message Passing for Efficient Active Inference Deployment	Nikola Pižurica, Matteo Rizzo, Nikola Milović, Alessio Burrello, Conor Heins, Míael de Prado Tim Verbelen, Toon Van de Maele Corrado Pezzato, Riddhi J. Pitliya
	31	Active Inference as the Reasoning layer for Physical AI	Guillem Monsó Wouter M. Kouw, Sepideh Adamiat
	37	Biological Bayesian Message Passing: A Brain-on-Chip with STDP-Based Training of hiPSC-Derived Neuronal Networks	Roberto Limongi, Angelica Maria Silva
	42	When Feedback Matters: An Active-Inference Model of the Speaking-Over-Writing Advantage in the Language-Production Effect	Stefano Ventre; Rodolfo Rossi Jialin Shi; Paolo Ossola Roberto Barumerli
	22	Active Inference of Social Exclusion: Computational Modelling of Cyberball in Borderline Personality Disorder	Nils Meyer-Kahlen, Lorenzo Picinali
	54	Framing Spatial Hearing within the Free Energy Principle: A Hierarchical Model of Perceptual Learning	



Posters



Poster Session	ID	Title	Authors
1. Neural Basis 14 Oct 16'00 - 17'00	116	A Tutorial on Predictive Coding Networks	<i>Björn van Zwol, Ro Jefferson, Egon L. van den Broek</i>
	61	Streaming Active Inference: A Problem Definition for Continuous Cognitive Runtimes	<i>Zihang Wang; Boris Sedlak</i>
	107	The universality class of hierarchical cognitive representation	<i>Adam Safron, Dalton A R Sakthivadivel</i>
	108	On the Nature(s) of Standing and Traveling Waves: How Brains Harmonically Self-Organize to Resonate with a Dynamic World	<i>Victoria Klimaj, Alex B Kiefer Dalton A R Sakthivadivel, Adam Safron</i>
	88	Putting the 'Body' in Embodied Cognition: An Active Inference Account of Concept Representation	<i>Daniel Revach</i>
2. Mah and Cog.Science 15 Oct 17'00 - 18'00	20	Amortizing Human Attention in Interactive Coding Agents: A Pilot Study	<i>Pengcheng Pan; Xinfang Zhang</i>
	41	Understanding Integrative Complexity Through Active Inference	<i>Dylan Grove; Alexander Hemming</i>
	68	The Rebundling Hypothesis: What LLMs Unbundle and Active Inference Could Restore	<i>Stefano Gorla</i>
	104	World Models: What They Are and What They Might Be	<i>Rudolf Mester</i>
	114	Mental rehearsal as active inference over modularized generative models	<i>Megumi Yoshihara; Takuya Isomura</i>
3. AI, Robotics, Applications 16 Oct 16'00 - 17'00	44	Lexical Specificity through Message Passing in Multiscale Active Inference	<i>Alexander Hemming; Dylan Grove</i>
	57	Belief Field Theory: Vector-Spin Collective Active Inference with Empirical Evidence	<i>Thomas P. Kehler</i>
	100	Active Inference for Intention-Aware Human–Robot Collaboration Under Uncertainty	<i>Shahriar Hassan; Pablo Bustos</i>
	119	Towards Adaptive Neuromorphic Sensing based on Active Efficient Coding	<i>Quentin Halbach; Vibhu Kaushik, Aheli Saha, Nguyen H. P. Nguyen, Arne Roennau, Chiara Bartolozzi, Alessio Franci, Elisabetta Chicca, René Schuster; Claudia Lenk</i>